JSON

LOGISTICS

- Furious Flying Fish due Friday, 3/20 @ 11:59pm
- HW6 (Podcast Feed) released tomorrow, due Wednesday 3/25 @ 11:59pm
- Midterm 2 on Monday, March 30th in class
- SRS every Sunday

ACTIVITY (FROM LAST TIME)

Hackers released a list of 10,000 passwords for PennKeys.

```
princess
abc123
666666
michael
abc123
adobeadobe
azerty
chocolate
753951
macromedia
121212
fdsa
...
```

You and your friends want to count how many common passwords are compromised. Your friends wrote a solution, but you want to make it run more efficiently. Modify the solution so that it is still correct (passing the unit tests) but so that it also runs at least 20% faster! There are a lot of ways you could do this, but experiment with:

- whether any computation being done here is unnecessary for the logic of the problem to be correct
- whether any choices of data structures could be substituted with others to speed things up.

```
def count_compromised(candidate_passwords: list[str],
                      hacked_filename: str) -> int:
```

Check Ed Lessons.

JSON & XML

JSON:

- Javascript Object Notation
- It's basically just Python dictionaries that get printed out. Convenient!
- Use the `json` library to read it.

Get Album

OAuth 2.0

Get Spotify catalog information for a single album.

Important policy notes

- Spotify content may not be downloaded
- Keep visual content in its original form
- Ensure content attribution

Request

GET /albums/{id}

id string Required

The Spotify ID of the album.

Example: `4aawyAB9vmqN3uQ7FjRGTy`

market string

cURL Wget HTTPie

```
curl --request GET \
  --url
https://api.spotify.com/v1/albums/4aawyAB9vmqN3uQ7FjRGTy \
  --header 'Authorization: Bearer
1POdFZRZbvb...qqillRxMr2z'
```

RESPONSE SAMPLE


```
{
  "album_type": "compilation",
  "total_tracks": 9,
  "available_markets": ["CA", "BR", "IT"],
  "external_urls": {
    "spotify": "string"
  },
  "href": "string",
  "id": "2up3OPMp9Tb4dAKM2erWXQ",
  "images": [
    {
      "url":
"https://i.scdn.co/image/ab67616d00001e02ff9ca10b55ce82ae553c8228",
      "height": 300,
      "width": 300
    }
  ],
  "name": "string",
  "release_date": "1981-12",
  "release_date_precision": "year",
  "restrictions": {
    "reason": "market"
  },
  "type": "album",
  "uri":
```

Data sent over the internet is usually complex and layered, so JSON is a good way of formatting it.

At right: JSON returned when querying Spotify for information about a particular album on its platform.

Can you see the different keys? (`album_type`, `total_tracks`, `id`, etc.)

- **RESPONSE SAMPLE**


```
{
  "album_type": "compilation",
  "total_tracks": 9,
  "available_markets": ["CA", "BR", "IT"],
  "external_urls": {
    "spotify": "string"
  },
  "href": "string",
  "id": "2up3OPMp9Tb4dAKM2erWXQ",
  "images": [
    {
      "url": "https://i.scdn.co/image/ab67616d00001e02ff9ca10b55ce82ae553c8228",
      "height": 300,
      "width": 300
    }
  ],
  "name": "string",
  "release_date": "1981-12",
  "release_date_precision": "year",
```

QUICK TIPS

JSON is just a series of lists, dictionaries, and primitive data types. *NO* sets, objects, or functions.

Get a JSON object out of a file and into your program with:

```
file = open("your_filename.json", "r")
my_data = json.load(file)
```

Once you've done the above, `my_data` will be a list or a dictionary containing the values it looks like it will contain.

```
[
 {'name' : 'CIS1100',
    'days' : ['M', 'W', 'F'],
    'time' : '12:00pm',
    'instructors' : [
        {'name' : 'Harry', 'dept' : 'CIS'},
        {'name' : 'Talie', 'dept' : 'CIS'}
    ]
 },
 {'name' : 'CIS1200',
    'days' : ['M', 'W', 'F'],
    'time' : '10:15am',
    'instructors' : [{'name' : 'Steve', 'dept' : 'CIS'}]
 },
]
```

How many courses are represented? If we parse this JSON into a list `l`, can you write an expression that produces that number of courses?

```
[
 {'name' : 'CIS1100',
    'days' : ['M', 'W', 'F'],
    'time' : '12:00pm',
    'instructors' : [
        {'name' : 'Harry', 'dept' : 'CIS'},
        {'name' : 'Talie', 'dept' : 'CIS'}
    ]
 },
 {'name' : 'CIS1200',
    'days' : ['M', 'W', 'F'],
    'time' : '10:15am',
    'instructors' : [{'name' : 'Steve', 'dept' : 'CIS'}]
 },
]
```

S7 What time does CIS 1100 meet? If we parse this JSON into a list `l`, can you write an expression that produces that value?

```
[
 {'name' : 'CIS1100',
    'days' : ['M', 'W', 'F'],
    'time' : '12:00pm',
    'instructors' : [
        {'name' : 'Harry', 'dept' : 'CIS'},
        {'name' : 'Talie', 'dept' : 'CIS'}
    ]
 },
 {'name' : 'CIS1200',
    'days' : ['M', 'W', 'F'],
    'time' : '10:15am',
    'instructors' : [{'name' : 'Steve', 'dept' : 'CIS'}]
 },
]
```

L11 Take some time to describe the structure of this JSON object. What keys do the upper level dictionaries have? The lower level dictionaries?

COMPLETE THE PROGRAM

Using your answer to L11, finish the snippet below so that it prints out a set of every instructor's name. **C12**

- Don't assume you know how many courses there are!
- Don't assume you know how many instructors each course has!

```
course_list = json.load(json_file_of_courses) # load the prev. JSON
```

EXPAND THE JSON

Using your answer to L11, write a snippet of code that would add your favorite other class to the list in the right structure. **C14**

```
course_list = json.load(json_file_of_courses) # load the prev. JSON
```